

Machine Learning Algorithms Laboratory

Experiment 5 Report

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study 1
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Submission Date	

1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2 Theory

2.1 Decision Tree Classifier

A Decision Tree recursively splits the feature space using impurity measures like the Gini Index or Entropy to form simple decision rules. As the tree grows deeper it can start fitting the noise in the training data instead of the actual pattern, so a Decision Tree is prone to high variance and overfitting if not controlled using parameters like max depth.

2.2 Random Forest Classifier

Random Forest is an ensemble method that builds many decision trees on different bootstrapped samples of the data (bagging), and at each split only a random subset of features is considered. The final prediction is obtained by majority voting across all the trees. Because the trees are decorrelated from each other (due to bootstrapping and random feature selection), the ensemble has lower variance than a single deep decision tree, which usually leads to better generalization.

3 Dataset Description

Dataset	Description
Dataset Name	Wisconsin Diagnostic Breast Cancer Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30 numerical features
Target Classes	Benign (0), Malignant (1)
Missing Values	None
Class Distribution	Benign: 357 (62.7%), Malignant: 212 (37.3%)
Train-Test Split	80% Train (455 samples), 20% Test (114 samples)

4 Exploratory Data Analysis

The diagnosis column originally had "B" and "M" labels, which were mapped to 0 (Benign) and 1 (Malignant) before training.

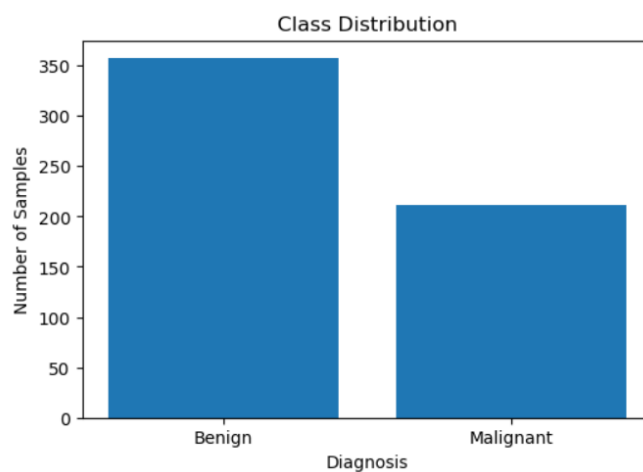


Figure 1: Class Distribution

Inference: The dataset is somewhat imbalanced, with about 63% benign and 37% malignant cases. Since there is some imbalance, precision and recall were checked along with accuracy, especially recall for the malignant class since missing a malignant case is a costlier mistake than a false alarm.

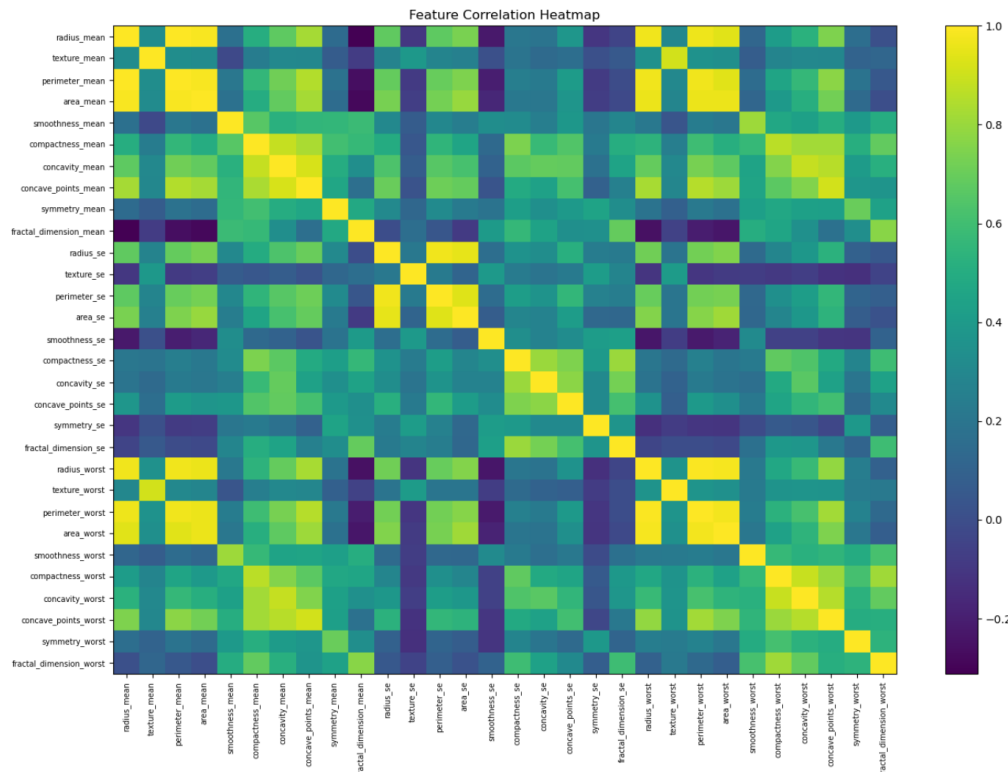


Figure 2: Feature Correlation Heatmap

Inference: Many of the features are highly correlated with each other, which makes sense since features like radius, perimeter and area of a cell nucleus are all related to its size, and the mean, standard error and worst value of each measurement are also naturally related. This kind of multicollinearity is not really an issue for tree based models like Decision Tree and Random Forest since they split on one feature at a time rather than using a linear combination of features.

5 Model Training and Hyperparameter Tuning

5.1 Decision Tree

A Decision Tree was tuned using GridSearchCV with 5-fold stratified cross validation over the following search space: criterion (gini, entropy), max_depth (2 to 10 and None), min_samples_split (2, 5, 10) and min_samples_leaf (1, 2, 4).

```
dt_param_grid = {
    "criterion": ["gini", "entropy"],
    "max_depth": [2, 3, 4, 5, 6, 8, 10, None],
    "min_samples_split": [2, 5, 10],
    "min_samples_leaf": [1, 2, 4]
}

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

dt_grid = GridSearchCV(DecisionTreeClassifier(random_state=42),
    dt_param_grid,
```

```

cv=cv, scoring={"accuracy": "accuracy", "
               f1": "f1"},
refit="accuracy")
dt_grid.fit(X_train, y_train)

```

The best parameters found were criterion=entropy, max_depth=4, min_samples_leaf=2, min_samples_split=2, giving a best CV accuracy of 0.9407.

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score (%)
entropy	4	94.07	91.63
entropy	4	94.07	91.63
entropy	4	93.85	91.42
entropy	4	93.85	91.45
entropy	4	93.63	91.08

Table 2: Decision Tree Hyperparameter Evaluation (Top 5 combinations)

Inference: All of the top performing combinations use entropy as the criterion and a fairly shallow max_depth of 4. This means a small tree (only 4 levels deep) is already enough to capture most of the pattern in this dataset, and deeper trees do not necessarily perform better on cross validation.

5.2 Random Forest

Random Forest was tuned in the same way, with a search space of n_estimators (50, 100, 200), max_depth (None, 5, 10, 15), max_features (sqrt, log2) and bootstrap (True, False).

```

rf_param_grid = {
    "n_estimators": [50, 100, 200],
    "max_depth": [None, 5, 10, 15],
    "max_features": ["sqrt", "log2"],
    "bootstrap": [True, False]
}

rf_grid = GridSearchCV(RandomForestClassifier(random_state=42),
    rf_param_grid,
                        cv=cv, scoring={"accuracy": "accuracy", "
                                         f1": "f1"},
                        refit="accuracy")
rf_grid.fit(X_train, y_train)

```

The best parameters found were bootstrap=False, max_depth=None, max_features=sqrt, n_estimators=200, giving a best CV accuracy of 0.9714.

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score (%)
200	5	sqrt	97.14	—
50	10	sqrt	97.14	—
200	10	sqrt	97.14	—
200	15	sqrt	97.14	—
200	None	sqrt	97.14	—

Table 3: Random Forest Hyperparameter Evaluation (Top 5 combinations)

Inference: Several different combinations of `n_estimators` and `max_depth` all reach the same top accuracy of 97.14%, and all of them use `max_features=sqrt`. This shows that once the number of trees is reasonably large (50 or more), the exact tree depth and number of estimators do not matter as much, but restricting the number of features considered at each split (`sqrt`) consistently helps.

6 Effect of Tree Depth on Overfitting

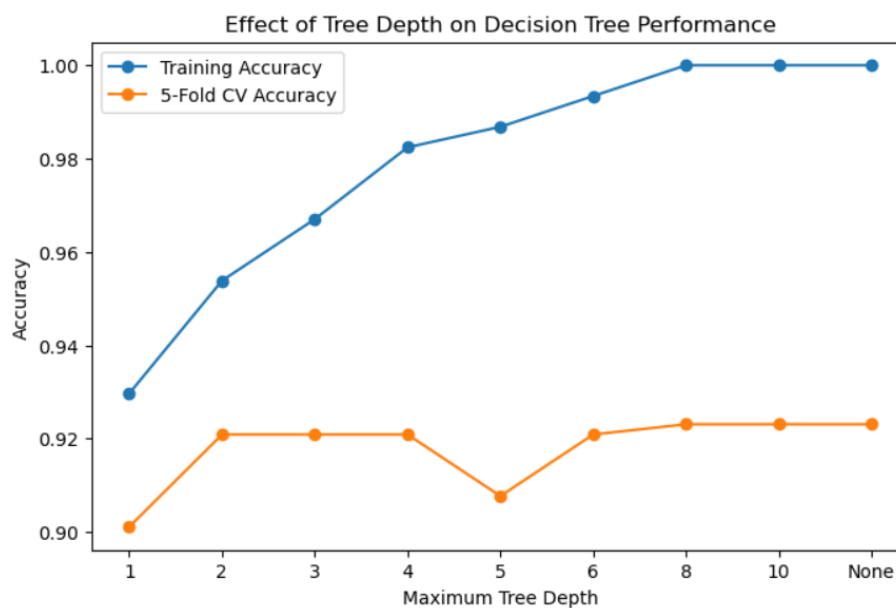


Figure 3: Effect of Maximum Tree Depth on Decision Tree Performance

Inference: As the max depth increases, the training accuracy keeps increasing and reaches close to 100% for unrestricted depth (None), while the cross-validation accuracy increases initially but then flattens out and even slightly decreases at very high depths. This gap between training and CV accuracy at higher depths is a clear sign of overfitting, the tree starts memorizing the training data instead of learning a generalizable pattern. This is exactly why the grid search picked a fairly small `max_depth` of 4 as the best option.

7 Evaluation on Test Set

Model	Accuracy	Precision	Recall	F1 Score	AUC
Decision Tree	92.98%	100.0%	80.95%	89.47%	95.63%
Random Forest	97.37%	100.0%	92.86%	96.30%	99.40%

Table 4: Test Set Performance Comparison

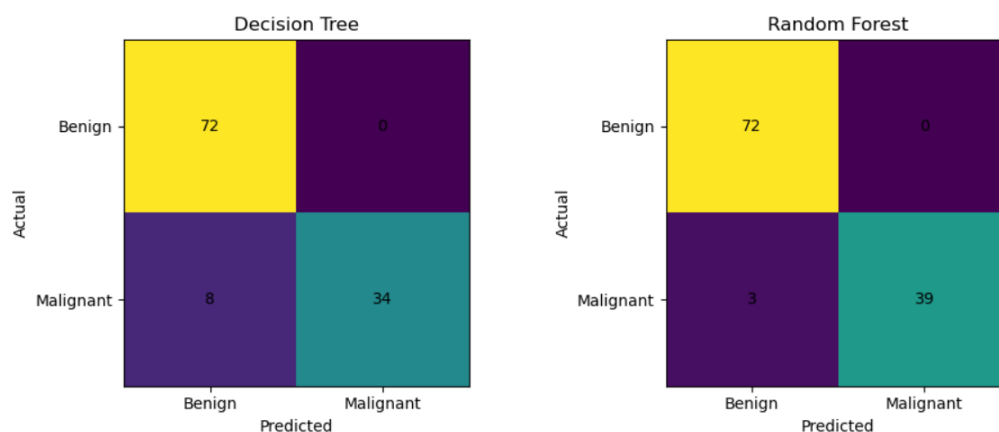


Figure 4: Confusion Matrix Comparison

Inference: Both models achieve perfect precision (100%), meaning neither model ever predicted a benign case as malignant on this test set. However, the Decision Tree missed 8 malignant cases (predicted as benign) while Random Forest missed only 3. Since missing a malignant case (false negative) is a much more serious error in a medical context than a false alarm, Random Forest’s higher recall (92.86% vs 80.95%) makes it the clearly safer model here.

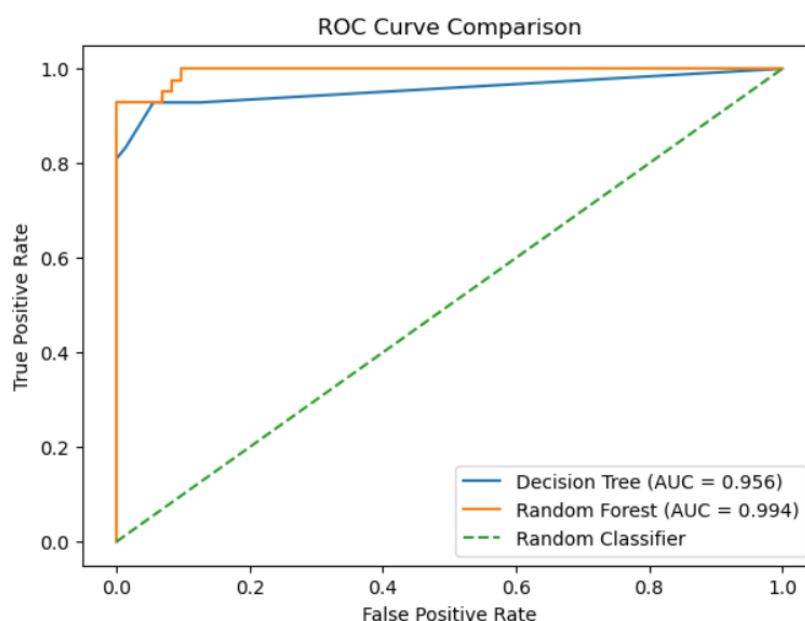


Figure 5: ROC Curve Comparison

Inference: Random Forest has a near-perfect ROC curve with an AUC of 0.994, compared to 0.956 for the Decision Tree. This confirms that Random Forest is better at ranking and separating the two classes across all classification thresholds, not just at the default 0.5 threshold.

8 5-Fold Cross-Validation Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	93.41	95.60	90.11	95.60	95.60	94.07
Random Forest	96.70	97.80	95.60	96.70	98.90	97.14

Table 5: 5-Fold Cross Validation Accuracy (%)

Inference: Random Forest is consistently better than the Decision Tree across every single fold, and its fold-to-fold variation is also smaller (ranging from about 95.6% to 98.9%) compared to the Decision Tree (90.1% to 95.6%). This lower variance across folds is exactly the kind of stability that ensemble methods like Random Forest are expected to provide.

9 Observation Questions

1. How does tree depth affect overfitting in Decision Trees?

As shown in the depth vs accuracy plot, increasing the max depth keeps improving the training accuracy (up to nearly 100%) but the cross-validation accuracy stops improving after a certain depth and can even drop slightly. A very deep tree starts fitting individual training samples/noise rather than the general pattern, which is a classic sign of overfitting. Limiting the depth (a max_depth of 4 was found to be best here) helps keep the model generalizable.

2. Which hyperparameter had the greatest impact on performance?

For the Decision Tree, max_depth had the biggest impact since all of the top performing combinations used the same shallow depth of 4 regardless of the min_samples_split/leaf values. For Random Forest, max_features (sqrt vs log2) seemed to matter the most, since every single top-performing combination used max_features=sqrt, while n_estimators and max_depth varied more freely among the top results.

3. How does Random Forest improve generalization?

Random Forest builds many decision trees on different bootstrapped samples of the data and also randomly restricts the features considered at each split. This makes the individual trees less correlated with each other, so when their predictions are combined using majority voting, the errors of individual trees tend to cancel out. This is why Random Forest achieved a higher and more stable cross-validation accuracy (97.14% average, smaller fold-to-fold variation) compared to the single Decision Tree (94.07% average).

4. Did ensemble learning always improve performance? Why or why not?

In this experiment, yes, Random Forest improved every single metric compared to the Decision Tree, accuracy, recall, F1-score and AUC were all higher, and cross-validation was also more stable. This is not guaranteed in every situation though, ensemble learning

generally helps most when the base model (a single decision tree here) has high variance to begin with, since bagging is specifically effective at reducing variance. If the base model already had very low variance and the dataset was very small, the benefit of an ensemble could be smaller, and it always comes at the cost of more training time and lower interpretability compared to a single tree.

10 Conclusion

In this experiment, a Decision Tree and a Random Forest were trained to classify breast cancer tumors as benign or malignant, with hyperparameters selected using 5-fold cross validation. The Decision Tree performed reasonably well but showed signs of overfitting at higher depths, and its cross-validation accuracy (94.07%) and test performance were both lower than Random Forest. Random Forest, by combining many decorrelated trees through bagging and random feature selection, achieved a higher and more stable accuracy (97.14% CV, 97.37% test) along with a much better recall for the malignant class, which is especially important in a medical diagnosis setting. This experiment clearly demonstrated how ensemble methods can reduce the variance and overfitting seen in a single decision tree.

11 References

1. Scikit-learn Documentation: Decision Trees. <https://scikit-learn.org/stable/modules/tree.html>
2. Scikit-learn Documentation: Random Forest. <https://scikit-learn.org/stable/modules/ensemble.html#forest>
3. Wisconsin Diagnostic Breast Cancer Dataset, UCI Machine Learning Repository.